



INDEPENDENT STUDIES

THIRD SESSION

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3,130 Corporate Finance

Major in Business Administration (BWL)

7. Capital Structure (BMAE, Chapters 16, 17, and 18)

7.1. True / False Questions

- a. Modigliani and Miller postulate that with perfect capital markets and when accounting for taxes the value of the firm is independent of its capital structure.
 - a. True
 - b. False**

- b. As the firm borrows more, a larger part of the firm's business risk is transferred from stockholders to bondholders.
 - a. True**
 - b. False

- c. Higher leverage always increases expected equity return, risk, and shareholder value.
 - a. True
 - b. False**

- d. Starting from a Modigliani and Miller world, the trade-off theory relaxes the assumptions of no taxes and no costs of financial distress.
 - a. True**
 - b. False

- e. In a Modigliani and Miller world, an increase in leverage results only in a lower return on equity when the return on assets exceeds the cost of debt.
 - a. True
 - b. False**

- f. According to the trade-off theory, debt finance has both benefits (tax shield) and costs (expected costs of financial distress). The marginal costs continually decrease the more debt a firm takes on such that the benefits outweigh the costs with increasing debt ratios.
 - a. True
 - b. False**

- g. The present value of the tax shield is the difference between the value of a firm under the assumption of full equity financing and the value of the same firm with some debt financing, accounting for the taxes in scenarios.
 - a. True**
 - b. False

- h. The pecking order theory explains the inverse (intra-industry) relationship between profitability and financial leverage since profitable firms depend less on external financing and possible tax shields are of secondary importance.

a. **True**
b. False

- i. The equity value reflects the value of the entire, i.e., firm to all investors.

a. True
b. **False**

- j. Firms are less likely to profit from an interest tax shield, the higher the probability to incur losses. Therefore, firms with low or highly fluctuating profits are expected to borrow less.

a. **True**
b. False

- k. Financial slack is most valuable to firms with few investment opportunities and generally poor prospects.

a. True
b. **False**

7.2. Tax shields

What are the PVs of the interest tax shields generated by these three debt issues if the marginal tax rate is 35%?

7.2.1.

The tax shield of a USD 1,000 perpetuity at 5% is most likely...

- a. ...USD 35.33
- b. ...USD 330.80
- c. ...USD 333.33
- d. ...USD 350.00**
- e. ...USD 363.33

$$PV(\text{tax shield}) = \frac{1,000 \cdot 0.05 \cdot 0.35}{0.05} = 350$$

7.2.2.

The tax shield of a five-year loan of USD 1,000 at 8% and no principal repayment until maturity is most likely...

- a. ...USD 25.97
- b. ...USD 111.80**
- c. ...USD 324.13
- d. ...USD 644.14
- e. ...USD 925.92

$$PV(\text{tax shield}) = \frac{0.35 \cdot 0.08 \cdot 1,000}{1 + 0.08} + \frac{28}{(1 + 0.08)^2} + \frac{28}{(1 + 0.08)^3} + \frac{28}{(1 + 0.08)^4} + \frac{28}{(1 + 0.08)^5}$$

$$PV(\text{tax shield}) = 111.80$$

7.2.3.

The tax shield of a USD 1,000 one-year loan at 6% is most likely...

- a. ...USD 19.81**
- b. ...USD 33.02
- c. ...USD 35.00
- d. ...USD 56.60
- e. ...USD 60.00

$$PV(\text{tax shield}) = \frac{0.35 \cdot 0.06 \cdot 1,000}{1 + 0.06} = 19.81$$

7.3. Tax shield of depreciations

If depreciation is USD 600,000 in the current period and the marginal tax rate is 35%, then the tax shield due to depreciation is most likely:

- a. **USD 210,000**
- b. USD 350,000
- c. USD 390,000
- d. USD 600,000
- e. Cannot be determined from the information given.

$$TS = 600,000 \cdot 0.35 = 210,000$$

7.4. Agency costs

Market values of Pear Corp. are illustrated in the following balance sheet:

Market values			
Net working capital	190	Bonds outstanding	260
Long-term assets	110	Common stock	40
<i>Total</i>	<i>300</i>	<i>Total</i>	<i>300</i>

Who wins and/or loses in the following situations?

7.4.1. Special dividend

Pear scrapes up USD 30 in cash and pays a cash dividend?

- a. **Stockholders win and bondholders lose**
- b. Bondholders win and stockholders lose
- c. Both win
- d. Both lose
- e. None of the answers a. to d. are correct.

There is less equity in the firm to bear losses and market risk.

7.4.2. Asset substitution

Pear discontinues its operation, sells all its fixed assets, and converts net working capital into USD 190. Unfortunately, the long-term assets fetch only USD 80 on the second-hand market. The USD 270 is invested in treasury bills.

- a. Stockholders win
- b. Bondholders win**
- c. Both win
- d. Both lose
- e. None of the answers a. to d. are correct.

Debt becomes risk-free since treasury bills can be assumed to be almost risk-free. Equity holders face a loss of 30 (i.e., equity value is down to 10) and they lose their upside potential.

7.5. Relevering of a firm

Firm ABC's capital structure consists of 30% debt and 70% equity.

7.5.1. Initial asset beta

What is the most likely asset beta for firm ABC, for an observed debt beta of 0.1 and an equity beta of 1.2?

- a. 0.70
- b. 0.80
- c. 0.84
- d. 0.87**
- e. 0.90

$$\beta_A = 0.1 \cdot 0.3 + 1.2 \cdot 0.7 = 0.87$$

7.5.2. New equity beta

Assuming a Modigliani and Miller world, what would be the most likely equity beta if the debt ratio was lowered to 25% and the debt beta remains constant?

- a. 0.93
- b. 1.00
- c. 1.05
- d. 1.13**
- e. 1.19

$$\begin{aligned}\beta_{E(25\%)} &= \beta_{Asset} + \frac{D}{E}(\beta_{Asset} - \beta_{Debt}) \\ \beta_{E(25\%)} &= 0.87 + \frac{0.25}{0.75}(0.87 - 0.1) = 1.1267 = 1.13\end{aligned}$$

7.5.3. New asset beta

Assuming a Modigliani and Miller world, what would the asset beta be if the debt ratio was lowered to 25% and the debt beta remains constant?

- a. 0.80
- b. 0.83
- c. 0.87**
- d. 0.90
- e. 0.92

$$\beta_A = 0.1 \cdot 0.25 + \left(0.87 + \frac{0.25}{0.75} (0.87 - 0.1) \right) \cdot 0.75 = 0.025 + 0.845 = 0.87$$

7.5.4. New asset beta

Assuming a Modigliani and Miller world, what would the asset beta be if the debt ratio was lowered to 25% and the debt beta lowers to 0.05?

- a. 0.80
- b. 0.83
- c. 0.87**
- d. 0.90
- e. 0.92

$$\beta_A = 0.05 \cdot 0.25 + \left(0.87 + \frac{0.25}{0.75} (0.87 - 0.05) \right) \cdot 0.75 = 0.0125 + 0.8575 = 0.87$$

The asset beta is unaffected under the assumptions of Modigliani & Miller.

7.6. Leverage

Cucumber Corp. has no debt outstanding. Under all-equity financing, the expected return on the company's shares is 15%. What would be Cucumber's after-tax WACC if the company issued debt and repurchased shares so that the D/V ratio would be 30%? Assume that the borrowing rate is 6% and the tax rate 35%.

- a. 11.72%
- b. 12.34%
- c. 14.37%**
- d. 15.00%
- e. 15.12%

$$WACC_{after-tax} = 0.3 \cdot 0.06 \cdot (1 - 0.35) + 0.7 \cdot \left(0.15 + \frac{0.3}{0.7} (0.15 - 0.06) \right) = 0.1437$$

Important: The cost of equity is given only under all equity financing. This changes when debt is added, and this had to be accounted for here

7.7. Indirect costs of bankruptcy

Which example is not attributed to indirect costs of bankruptcy?

- a. Managerial attention diverted from managing assets to managing liabilities.
- b. Firm loses flexibility when monitored closely by creditors.
- c. Assets must be sold in fire-sales.
- d. The firm has to pay expenses for lawyers in bankruptcy.**
- e. Intangible assets may be destroyed if the firm is broken up or sold.

8. Risk Management (BMAE, Chapters 27 and 28)

8.1. True / False Questions

- a. In the real world with market frictions, hedging can lower costs of financial distress so that hedging increases firm value.
 - a. **True**
 - b. False
- b. Hedging can reduce the volatility of a firm's cash flows. Therefore, the planning of CAPEX becomes more reliable, increasing firm value.
 - a. **True**
 - b. False
- c. Buying a futures contract implies paying now for delivery at a future date.
 - a. True
 - b. **False**
- d. The impact of several risk factors on the volatility of a firm's cash flows can be estimated by estimating a regression with historical data, if one expects that the firm's exposure to these risk factors remains constant.
 - a. **True**
 - b. False
- e. To apply a historical Value-at-Risk, you have to assume a historical distribution of the returns.
 - a. True
 - b. **False**
- f. A left-skewed distribution faces more extreme positive than negative events.
 - a. True
 - b. **False**
- g. The expected shortfall only considers the frequency of large losses but neglects the magnitude of these losses.
 - a. True
 - b. **False**
- h. When you buy a future (long position), you gain if the price of the underlying exceeds the future price at time T.
 - a. **True**
 - b. False

- i. Academic evidence (e.g., Nance, Smith, and Smithson, 1993) shows that larger firms hardly use derivatives for hedging purposes since they have enough cash to bear the risk. This allows large firms to save the costs of hedging.
 - a. True
 - b. False**

- j. The holder of a commodities futures contract does not have to pay for storage costs, but foregoes the convenience yield.
 - a. True**
 - b. False

- k. Academic evidence (e.g., Graham and Rogers, 2002) suggests that firms which hedge more have higher debt ratios.
 - a. True**
 - b. False

- l. Academic evidence (e.g., Nance, Smith, and Smithson, 1993) suggests that a lower dividend payout is associated with a lower probability of corporate hedging. This was to be expected as these firms are not “obliged” to support a dividend and thus can accommodate more volatile cash flows.
 - a. True**
 - b. False

8.2. Factor-based volatility of a firm's cash flows

Assume that we obtained the following results from estimating a two-factor model:

$$CF = 25 + 3F_{curr} - 6F_{int}$$

CF : Cash flow

F_{curr} : percentage change in the ¥/USD exchange rate over the next year

F_{int} : percentage change in three-month LIBOR from now until one year from now

The expected variance / covariance structure of the factor exposures is:

$$\sigma_{curr}^2 = 0.021 \quad \sigma_{int}^2 = 0.032 \quad cov_{curr/int} = 0.006$$

What is the most likely factor-based volatility of the firm's cash flows?

- a. 0.000
- b. 0.006
- c. 0.255
- d. 1.060**
- e. 1.340

$$\sigma_{CF} = \sqrt{3^2 \cdot 0.021 + 6^2 \cdot 0.032 + 2 \cdot 3 \cdot (-6) \cdot 0.006} = 1.06$$

8.3. Parametric Value-at-Risk

Consider the following information:

- Investment volume = USD 1,000,000
- Daily volatility: $\sigma_{Daily} = 2.5\%$
- Daily returns are normally distributed

What is the most likely 2.5% Value-at-Risk (VaR) of this investment over a 16-day period?

- USD 25,000
- USD 49,000
- USD 100,000
- USD 196,000**
- USD 250,000

$$VaR(2.5\%)^{16} = 1,000,000 \cdot 1.96 \cdot 0.025 \cdot \sqrt{16} = 196,000$$

8.4. Expected shortfall

Probability of state	Payoff per state	Investment	Profit
10%	0	-40	-40
20%	10	-40	-30
40%	50	-40	10
30%	80	-40	40

What is the most likely expected shortfall at a 40%-level?

- 40.0
- 33.3
- 22.5**
- 10.0
- 40.0

$$ES_{40\%} = \frac{0.1 \cdot (-40) + 0.2 \cdot (-30) + 0.1 \cdot 10}{0.40} = -22.5$$

8.5. Money market hedge

Assume that the exchange rate between CHF/USD is 1:1.2. The risk-free interest rate in Switzerland is 2% p.a. The risk-free interest rate in the US is 5% p.a. What is the most likely current two-year future price F_2 ?

- a. 0.79
- b. 1.27**
- c. 1.32
- d. 1.37
- e. 1.42

$$F_2 = \frac{1.2 \cdot (1 + 0.05)^2}{(1 + 0.02)^2} = 1.27$$

8.6. Currency hedge

Beta Barrels Inc. is a U.S.-based company considering the construction of a new barrel plant in Spain. The forecasted cash flows in millions of Euros are as follows:

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
- 80	+10	+20	+23	+27	+25

The spot exchange rate is USD 1.2 = 1 EUR. The interest rate in the U.S. is 8% and the Euro interest rate is 6%. We assume that the barrel production is effectively risk-free.

What is the NPV of the dollar cash flows from the project, if the company hedges against exchange rate changes?

- a. USD 1.26 million
- b. USD 3.47 million
- c. USD 4.52 million
- d. USD 6.13 million
- e. USD 7.93 million**

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Forward rates	1.2	$1.2 \cdot \frac{1.08}{1.06} = 1.2226$	$1.2226 \cdot \frac{1.08}{1.06} = 1.2457$	$1.2457 \cdot \frac{1.08}{1.06} = 1.2692$	$1.2692 \cdot \frac{1.08}{1.06} = 1.2931$	$1.2931 \cdot \frac{1.08}{1.06} = 1.3175$
Dollar cash flows	$-80 \cdot 1.2 = -96$	$10 \cdot 1.2226 = 12.23$	$20 \cdot 1.2457 = 24.91$	$23 \cdot 1.2692 = 29.19$	$27 \cdot 1.2931 = 34.91$	$25 \cdot 1.3175 = 32.94$

$$NPV = -96 + \frac{12.23}{(1.08)^1} + \frac{24.91}{(1.08)^2} + \frac{29.19}{(1.08)^3} + \frac{34.91}{(1.08)^4} + \frac{32.94}{(1.08)^5} = 7.93$$

8.7. Basis risk

In which of the following cases would you expect basis risk to be serious?

- I) A broker owning a large block of Disney common stock hedges by selling index futures;
- II) A corn farmer in Iowa hedges the selling price of her crop by selling corn futures at the Chicago Board Options Exchange;
- III) A U.S.-based importer must pay 900 million Euros in six months and she hedges by buying Euros forward.

Which of the following statements is correct?

- a. I) involves limited basis risk, II) significant basis risk, and III) no basis risk
 - b. I) involves limited basis risk, II) no basis risk, and III) significant basis risk
 - c. I) involves significant basis risk, II) no basis risk, and III) limited basis risk
 - d. I) involves a significant basis risk, II) limited basis risk, and III) no basis risk**
 - e. I) involves no basis risk, II) significant basis risk, and III) limited basis risk
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- *Disney stock is imperfectly correlated with the market, imposing significant basis risk.*
 - *Value of farmer's corn may be highly but not perfectly correlated with that of the corn in the futures contract, limited basis risk.*
 - *Hedging currency risk involves no basis risk.*